

English
Medium



EDU TERIA

72nd BPSC

Prelims Online Batch

BIOLOGY

LECTURE

12

NUTRITION-Part-03



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Fat

Fat, in nutritional and biological terms, describes a type of macronutrient and lipid that serves as a concentrated source of energy, providing roughly 9 calories per gram—more than double that of carbohydrates or proteins.

- Fats are organic compounds and essential macronutrients composed of carbon, hydrogen, and oxygen, and they belong to the lipid family.
- **Energy Source:** Fats are the most concentrated source of energy, providing about 9 calories per gram, which is more than double the 4 calories per gram provided by carbohydrates or proteins.
- **Storage and Insulation:** The body stores excess calories as fat in adipose tissue, which provides energy reserves and helps insulate the body and cushion vital organs.
- **Vitamin Absorption:** Fats are crucial for the absorption and transport of fat-soluble vitamins (A, D, E, and K) through bloodstream.
- **Cell Structure and Hormones:** Lipids are essential components of cell membranes and are also involved in the production of certain hormones.
- **Essential Fatty Acids:** Some fatty acids are "essential" because the body cannot synthesize them and must obtain them through diet.

Types of Fats and Their Sources:

- Fats are categorized based on their chemical structure (presence or absence of double bonds) and their effects on health

Type of Fat	Health implication	Common source
Saturated Fats	Considered unhealthy in excess; can raise bad (LDL) cholesterol and increase heart disease risk.	Animal products like meat, butter, cheese, milk, and certain tropical oils (coconut, palm oil).
Trans Fats	The most harmful type of fat; artificially produced (partially hydrogenated oils) and strongly linked to heart disease.	Processed and packaged foods, fried foods, cookies, cakes, and some natural sources like meat and dairy in small amounts.
Monounsaturated Fats (MUFAs)	Healthy fats; help improve blood cholesterol levels and control inflammation.	Olive oil, avocado, peanuts, and nuts like almonds and hazelnuts.
Polyunsaturated Fats (PUFAs)	Healthy fats (includes Omega-3 and Omega-6 fatty acids); essential for brain development, heart health, and immune function.	Vegetable oils, walnuts, flax seeds, and oily fish like salmon and mackerel.

Saturated fatty acid

- (a) Only single bond, single bonds are present between C-C atoms
 - (b) Also known as non-essential fatty acid, because they can be synthesized in animal body, so it is not essential to take them with food.
 - (c) Lipids with more amount of saturated fatty acids are mostly present in solid form at normal temperature.
- e.g.** Butter, Ghee
- (d) Metabolically they are very less reactive. So mostly don't participate in metabolism. So have tendency to store in animal body and cause obesity. Some amount of saturated fatty acids convert into cholesterol in liver. This cholesterol deposit on inner wall of blood vessels, so cause high blood pressure & different types of cardio-vascular disease.
 - (e) Mostly animal fat contains high amount of saturated fatty acids,

Examples

	No. of C
Palmitic acid	16
Stearic acid	18

Unsaturated fatty acid

- (a) Double bonds also present with single bonds between C-Atoms.
 - (b) Also known as essential Fatty acid because they can't be synthesized in animal body, so it is essential to take them with food.
 - (c) Lipids with more amount of unsaturated fatty acids are mostly present in liquid form at normal temperature.
- e.g.** Coconut oil, Mustard oil.
- (d) Metabolically they are highly reactive, so mostly participate in metabolic reactions. So don't have tendency to store in animal body. So no obesity, no cholesterol formation, no cardiovascular disease, no high B.P. & no harmful effect.
 - (e) Mostly plant fat contains high amount of unsaturated fatty acids.

- (f) Example
- | | |
|--|---------------------|
| | No. of double bonds |
| Oleic acid = 1 MUFA/Monounsaturated fatty acid | |
| Linoleic acid = 2 PUFA/ Polyunsaturated fatty acid | |
| Linolenic acid = 3 | |
| Arachidonic acid = 4 | |



Vitamin

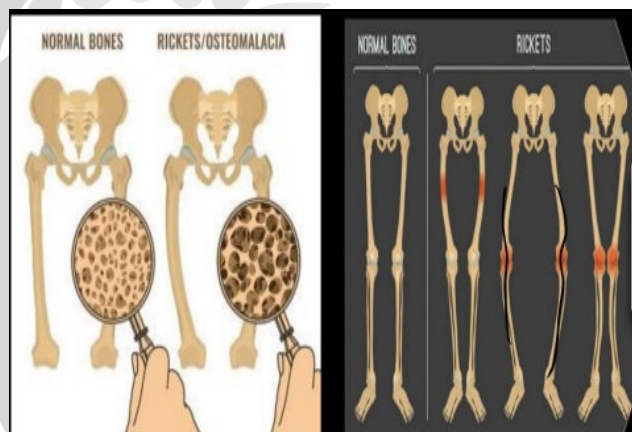
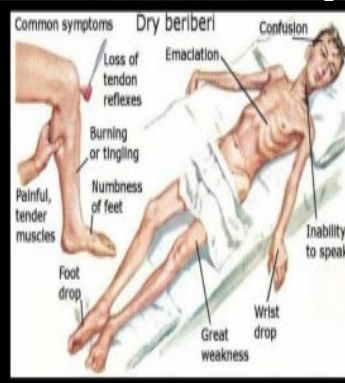
Vitamins are organic micronutrients required in small amounts for normal growth, metabolism, immunity, and proper functioning of the body. They do not provide energy but help regulate many biochemical processes. Their absence causes specific Vitamin Deficiency Diseases.

Vitamin	Common Name	Deficiency Disease
A	Retinol	Night blindness (Nyctalopia), Xerophthalmia, Dermatitis
B ₁	Thiamine	Beri-beri, Polyneuritis, Cardiovascular atrophy
B ₂	Riboflavin (Vitamin G)/ Yellow Enzyme	Cheilosis, Glossitis, Keratitis
B ₁₂	Cyano-Cobalamine	Pernicious anaemia (Megaloblastic anaemia)
C	Ascorbic Acid	Scurvy, Anaemia, Joint pain
D	Calciferol	Rickets in children and Osteomalacia in adults
E	Tocopherol / Ant sterility / Beauty Vitamin	Macrocytic anaemia, Muscular dystrophy
K	Menadione / Phylloquinone	Severe bleeding

Vitamin	Sources
Vitamin A	Liver, fish oils, eggs, milk, carrots, orange fruits, leafy greens
Vitamin C	Citrus fruits (oranges, grapefruit), strawberries, broccoli, peppers, tomatoes
Vitamin D	Sunlight exposure (skin synthesis), fortified milk, fatty fish, egg yolks
Vitamin E	Vegetable oils, nuts (almonds, hazelnuts), seeds (sunflower), spinach
Vitamin K	Leafy greens (spinach, kale), broccoli, soybeans, vegetable oils
Vitamin B1	(Thiamin) Pork, whole grains, brown rice, potatoes, legumes
Vitamin B2	(Riboflavin) Dairy products, meat, liver, green vegetables, refined grains
Vitamin B3	(Niacin) Meat, fish, poultry, eggs, nuts, wheat flour, legumes
Vitamin B5	Meat, poultry, fish, bananas, potatoes, vegetables
Vitamin B9	(Folate) Leafy greens, citrus fruits, beans, liver, fortified grains
Vitamin B12	Meat, fish, poultry, dairy products, eggs, fortified cereals



किलोसिस
Cheilosis



Vitamins and Minerals

- Micronutrients are essential dietary components required in small amounts for proper metabolism, growth, development, and immune function.
- Major Micronutrient Deficiencies
 - Iron deficiency → Anemia
 - Iodine deficiency → Goiter
 - Vitamin A deficiency → Night blindness

Classification of Micronutrients

1. Vitamins (Organic Compounds) Funk

A. Fat-Soluble Vitamins

Stored in body fat and liver.





- Vitamin A
- Vitamin D
- Vitamin E
- Vitamin K

B. Water-Soluble Vitamins

Not stored in large amounts; require regular intake.

- Vitamin B-complex (B1, B2, B3, B6, B9, B12 etc.)
- Vitamin C

2. Minerals (Inorganic Elements)

A. Microminerals (Needed in large amounts)

- Calcium
- Potassium
- Sodium
- Phosphorus
- Magnesium

B. Trace Minerals (Needed in small amounts)

- Iron
- Iodine
- Zinc
- Copper
- Fluoride

Minerals and Deficiency diseases

Mineral	Major Function	Deficiency Disease / Condition
Iron (Fe)	Oxygen transport (hemoglobin)	Anemia (fatigue, weakness)
Iodine(I)	Thyroid hormone synthesis	Goiter, hypothyroidism, cretinism
Zinc(Zn)	Wound healing, immunity	Skin lesions, impaired immunity
Calcium (Ca)	Bone development, blood clotting	Osteoporosis, rickets, tetany
Fluoride (F)	Dental health	Dental cavities
Copper (Cu)	Iron utilization, never function	Anemia, Neurological problem

Vitamins and Deficiency Diseases

Vitamin	Function	Deficiency Disease
Vitamin A	Vision skin health, immunity	Night blindness, Xerophthalmia
Vitamin D	Calcium absorption	Rickets (Children), Osteomalacia (adults)
Vitamin C	Collagen synthesis, immunity	Scurvy (bleeding gums)
Vitamin B1(Thiamine)	Energy metabolism	Beriberi
Vitamin B9 (Folate)/B11	DBA synthesis, cell vision	Megaloblastic anemia, birth defects
Vitamin B12	Red blood cell formation, nerve function	Pernicious anemia

